

Pathogen induced disturbance and succession in temperate forests: Evidence from a 100-year data set in southern Sweden



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Abstract

Major tree species are declining in many temperate forests due to changing disturbance regimes, including invasive pests and pathogens. We examined the interaction of secondary succession and Dutch elm disease in the Swedish temperate forest reserve Dalby Söderskog, based on five tree surveys made between 1909 and 2011. The forest is characterized by the coexistence of four major European tree species: wych elm (*Ulmus glabra*), European ash (*Fraxinus excelsior*), European beech (*Fagus sylvatica*) and pedunculate oak (*Quercus robur*). After protection of the forest in 1918, lack of disturbance mainly favoured elm, while the oak population declined due to mortality of old oaks and lack of regeneration. Dutch elm disease has caused high and continuous elm mortality since 1988. As a result, increased light availability at the forest floor favoured abundant regeneration of ash, beech, and lately also oak. The recent arrival of an invasive fungal pathogen causing ash dieback may once again change the course of succession. Open space emerging from loss of elm and ash in forest reserves may be used by reserve managers to favour oak regeneration and biodiversity of semi-open woodlands once lost during succession to closed forest. We conclude that winners and losers change places as an effect of invasive pathogens, resulting in unexpected successions and both losses and gains in valuable ecological niches and habitat structures in temperate broadleaf forests.

Zusammenfassung

Wichtige Baumarten sind derzeit in den Wäldern der gemäßigten Zonen stark rückläufig, vor allem aufgrund sich ändernder Störungsdynamik, einschließlich invasiver Schädlinge und Krankheitserreger. Wir untersuchten die Wechselwirkung von Sekundärsukzession und Ulmensterben in einem südschwedischen Naturwaldreservat (Dalby Söderskog), basierend auf fünf Bestandesinventuren zwischen 1909 und 2011. Der Wald wird durch die historische Koexistenz von Bergulme (*Ulmus glabra*), Esche (*Fraxinus excelsior*), Buche (*Fagus sylvatica*) und Stieleiche (*Quercus robur*) geprägt. Nach der Unterschutzstellung des Waldes im Jahr 1918 nahm vor allem die Ulme auf Kosten der sich nicht mehr verjüngenden Eiche zu. Seit 1988 hat jedoch das Ulmensterben den Ulmenbestand erheblich reduziert. Das dadurch wieder erhöhte Lichtangebot hat eine reichliche Verjüngung von Esche, Buche und zuletzt auch Eiche begünstigt. Die Ankunft des Eschentriebsterbens, einer neuartigen Pilzkrankheit, könnte allerdings erneut die Waldentwicklung in eine unvorhergesehene Richtung lenken. Der starke Rückgang der Ulme und

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eventuell auch der Esche bedeutet einen schwerwiegenden Verlust von Lebensraum, vor allem für epiphytische Moose und Flechten, die nur teilweise auf den verbleibenden Eichen und Buchen wachsen können. Große Bestandeslücken aufgrund von Ulmen- und Eschensterben sollten allerdings genutzt werden, um die Eichenverjüngung und die Entwicklung anderer wertvoller Habitatstrukturen halboffener Lebensräume zu fördern, die im Laufe des 20. Jahrhunderts in vielen geschützten Wäldern verloren gegangen sind.

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Introduction

Tree diversity in temperate forests depends on the long-term interactions of canopy disturbances and understorey recruitment. The current dynamics in unmanaged temperate forests are often characterized by a change from semi-open stand structures maintained by forest grazing or regular low intensity fires towards increasingly dense canopies (Abrams 2003; Vera 2000). This process may eventually result in the lack of regeneration by light-demanding tree species, such as oaks (Abrams 2003; McEwan, Dyer, & Pederson 2011).

However, since the early 20th century, many temperate forests are increasingly affected by severe disturbance from introduced pests and pathogens (Ellison, Bank, Clinton, Colburn, Elliott, & Ford 2005), although high-impact pathogen outbreaks may have occasionally occurred throughout the Holocene (Waller 2013). In North America these introductions have resulted in the decline of several important tree species (Lovett, Canham, Arthur, Weathers, & Fitzhugh 2006). In Europe, Dutch elm disease, caused by fungal pathogens (*Ophiostoma* spp.) and spread by bark beetles (*Scolytus* spp.), brought about a widespread population decline of several elm species (e.g. Peterken & Mountford 1998). Recently, also European ash (*Fraxinus excelsior*) has been affected by an emerging invasive fungal pathogen causing widespread ash dieback (Pautasso, Aas, Queloz, & Holdenrieder 2013).

A loss of tree species may have a major impact on forest ecosystem processes and biodiversity (Ellison et al. 2005). Exotic pests and pathogens often produce more long-lasting changes in species composition than wind or fire disturbances, because they can target specific tree species and remain as permanent components of the ecosystem (Lovett et al. 2006). However, death of dominant tree species by disease also results in canopy opening, and establishment of shade-intolerant woody and herbaceous species, which at least locally may increase total species diversity of the forest vegetation (Smith, Behrens, & Sax 2009).

Predicting ecological impacts requires a thorough understanding of the successional processes following changing disturbance regimes, and should be based on long-term ecological studies (Lindenmayer, Likens, Krebs, & Hobbs

2010; McEwan et al. 2011). Unfortunately, only a very limited number of long-term studies are available to provide detailed data over more than 50 years on changes in forest structure and composition. For the Dalby Söderskog reserve in southern Sweden presented in this study, an exceptionally long time series of surveys during 100 years is available to study changes in tree size distributions and canopy structure. In addition, this forest exemplifies the widespread succession of abandoned pasture woodland towards closed canopy hardwood forest (Vera 2000). Prior to protection in 1918, forest grazing and selective cuttings had resulted in a long-term coexistence of four major European tree species, maintaining a relatively high structural and functional diversity: wych elm (*Ulmus glabra*), European ash, European beech (*Fagus sylvatica*) and pedunculate oak (*Quercus robur*) (Lindquist 1938). Centuries of this wood pasture management were followed by 70 years of undisturbed forest succession before Dutch elm disease started to affect forest dynamics around 1988.

In this study, we aim to determine if the changes in forest structure and composition were consistent with the following hypotheses:

- (1) Lack of disturbance prevents regeneration of light-demanding species (ash, oak), while shade-tolerant species increase (elm, beech).
- (2) The impact of Dutch elm disease is size dependent, and primarily affects trees in upper and mid-canopy layers.
- (3) Increased light availability in gaps after Dutch elm disease results in new regeneration of ash and oak, and release of existing advanced regeneration of beech.
- (4) Long periods of undisturbed succession, as well as severe disturbance by invasive pests and pathogens result in large gaps in the size distributions of the affected species.

Materials and methods

Study area

Dalby Söderskog has a size of 37 ha and is situated in the county of Skåne, southern Sweden (55°41' N, 13°20' E,

50–75 m a.s.l.). The mineral soil, loamy clay with a mean pH (H₂O) of 6.1 (+ 0.08 SE, Persson, Malmer, & Wallén 1987), is derived from glacial till (Weichsel glacial period) and supports a vegetation of nutrient demanding forest herbs, e.g. *Anemone ranunculoides*, *Corydalis cava* and *Mercurialis perennis* in the ground layer, and hazel (*Corylus avellana*) and hawthorn (*Crataegus laevigata*) in the shrub layer (von Oheimb & Brunet 2007). The slopes along a small stream running through the southeastern part are relatively well-drained (Appendix: Fig. 1), while the major part of the forest has moist or wet soils (Lindquist 1938). The area has a temperate, sub-oceanic climate, with a mean annual temperature of 7.5 °C and a mean annual precipitation of ca 650 mm. Due to large amounts of old trees and coarse woody debris, the diversity of saproxylic insects and fungi, and of epiphytic lichens and bryophytes is high, including many taxa of conservation concern (County Administrative Board of Skåne, unpublished data).

Grazing ceased gradually during the 19th century and the last significant tree harvest was carried out in 1914–16 (ca 1600 m³ of wood cut out of a total stock of ca 8000 m³, Lindquist 1938). In general, most of the forest has experienced secondary forest succession without active management since then. Dead elm trees are felled since 1988 for safety reasons when standing close to hiking paths. Further details on management measures and other disturbances influencing the study area during the past 300 years can be found in Appendix A: Tables 1 and 3.

Previous surveys 1909–1970

In 1935, the ecologist Bertil Lindquist established a system of transect lines and sample plots to study the forest structure and vegetation (Lindquist 1938). He used a straight path through the forest as a base line and mapped trees along 16 transect lines perpendicular to the path, and in 74 sample plots located on these lines (see Appendix A: Fig. 1). For the present study, the grid of sample plots was reconstructed in 2010 using an aerial photograph of the forest with Lindquist' map as a digital overlay and the corresponding GPS coordinates. Published data on previous tree surveys were also available from 1909, 1916 (summarized in Lindquist 1938) and 1970 (Malmer, Lindgren, & Persson 1978, see Appendix A: Table 2). Available published data from 1909 only included species-wise total stem numbers (≥ 20 cm stem diameter, Lindquist 1938), while the data from 1916, 1935 and 1970 were published in 5 cm diameter classes (see Appendix A: Table 2 for further details). Data on the frequency of woody species in the shrub and ground layers in the 74 systematically distributed plots were available from surveys in 1935 (Lindquist 1938) and 1969 (Malmer et al. 1978).

Tree survey

In late 2011 and early 2012, the diameter at 1.3 m height (dbh) of all standing woody stems ≥ 20 cm dbh was measured

and identified to species in circular 314 m² (10 m radius) plots using the centre points of the 74 plots. In circular 100 m² plots (5.64 m radius, same centre point) all standing stems with a dbh of 10–19 cm were recorded (see Appendix A: Table 2). In addition, a complete inventory of the whole forest was done for oaks >1.3 m tree height and for living ash and beech trees with a dbh ≥ 100 cm (see Appendix A: Table 2).

All stems were classified as either living or dead. In addition, the dead oaks were recorded as standing (snag) or downed (log). Living elm and ash trees were classified as either healthy, or visibly damaged by Dutch elm disease or ash dieback. Elms were judged as damaged when epicormic sprouting, dying bark and dieback at the top of the tree was recorded, while ash was classified as damaged when a significant part of the tree crown was either dead or top shoots were severely stunted.

Survey of woody species in the shrub and ground layers

To assess the frequency and abundance of regeneration, the presence of woody species in the shrub and ground layers was recorded and percentage cover was estimated in the 74 plots in July 2010. The shrub layer was differentiated into two height classes (S1: 2–8 m and S2: 1–2 m). The ground layer included woody species <1 m in height. The plot size was 4 m $\times$ 4 m for the shrub layers, and 1 m $\times$ 1 m for the ground layer, with identical plot centres, following Lindquist' (1938) original method, and a similar re-survey in 1969 (see Appendix A: Table 2).

Data analysis

The data from the different tree surveys were standardized to stem density ha⁻¹ in different dbh classes. Lists of living trees were reconstructed for the surveys of 1916, 1935 and 1970 for elm, ash, beech and oak by using the mid-values of the 5 cm diameter classes. These tree lists were then used to perform tests of the diameter distributions, including our survey of 2011. Differences in diameter distributions between sampling occasions within tree species were tested for by two-sample Kolmogorov–Smirnov tests (KS-test). These tests were repeated for each pair ($n=6$) of sampling occasions within species. The KS-tests were done by applying the Stats package in R 2.15.0 (RDCT 2012). The KS-tests were done on data of trees per hectare.

The frequencies of elm, ash, beech and oak in the shrub and ground layers across all 74 plots were calculated and differences in species frequencies between 1970 and 2011 were tested with a χ^2 -test. Changes in proportions of vitality classes in relation to dbh classes were studied with linear regression.

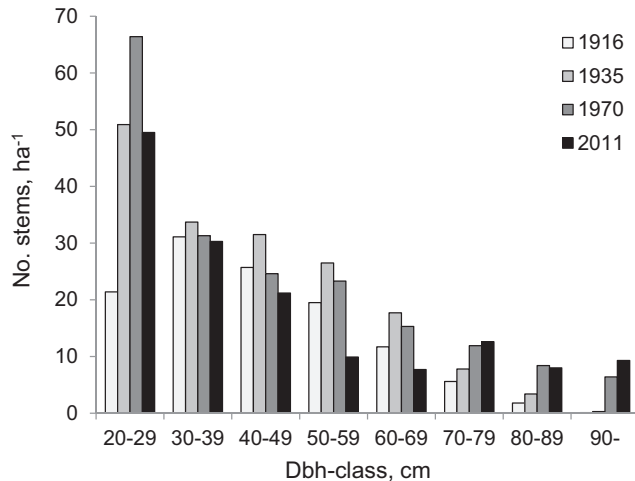


Fig. 1. Overall diameter distribution in Dalby Söderskog during the period 1916–2011 (no. of all stems ha^{-1}). Only living trees with stem diameter ≥ 20 cm at breast height are included.

Results

General changes in diameter structure and forest density

In 1916, mid-sized trees dominated and only two trees had a dbh larger than 90 cm (Fig. 1). In 1935, many younger trees had grown into the 20–29 cm dbh class. This development continued until 1970, when the number of large trees also had increased considerably. In 2011, however, a lower stem density than in 1970 was recorded in all dbh classes from 20 to 69 cm (Fig. 1). The density of very large trees (≥ 90 cm dbh), on the other hand had surpassed nine trees ha^{-1} . Along with these changes in size distribution, the basal area increased by more than 70% from 1916 to 1970, but decreased again by 16% from 1970 to 2011 (Fig. 2).

Elm

Elm peaked in 1970 as the most common species in the forest (≥ 20 cm dbh, Fig. 2), but by 2011, Dutch elm disease had greatly reduced the elm population. The development of the dbh distribution shows the increasing dominance of elm in the lower and intermediate dbh classes until 1970 (see Appendix A: Table 3), and the collapse of the population until 2011, except for trees smaller than 30 cm dbh (Fig. 3). In summary, there were only small changes in diameter distribution before Dutch elm disease, while the distribution in 2011 was significantly different from all the other years (Table 1). The proportion of dead trees increased significantly with increasing dbh, which confirmed hypotheses 2 and 4 (Fig. 4, $R^2 = 0.87$, $P = 0.006$, linear regression). However, in the upper shrub layer, elm still held the position of the most abundant species in 2011, having a mean cover of

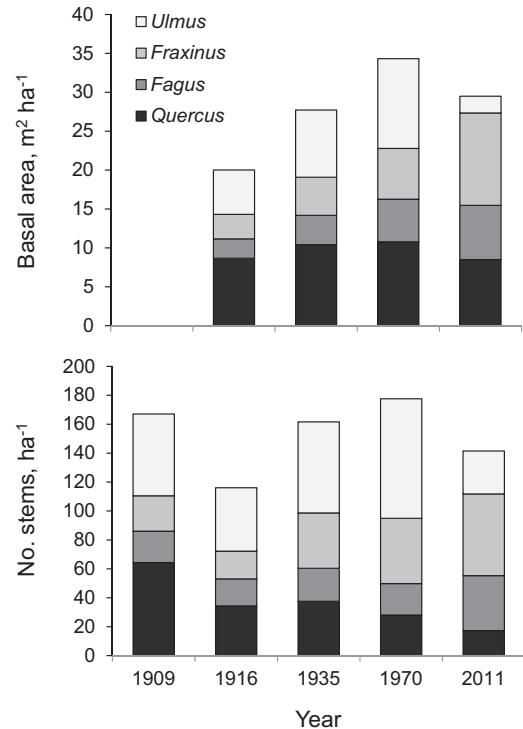


Fig. 2. Basal area and number of stems per hectare of the four main tree species in Dalby Söderskog according to surveys in 1909 (only number of stems), 1916, 1935, 1970 and 2011. Only living trees with stem diameter ≥ 20 cm at breast height are included.

25% and an overall frequency of 72% in this layer (Table 2, see also Appendix A: Table 3, for stems with 10–19 cm dbh).

Ash

The number of ash trees ≥ 20 cm dbh has increased continuously since 1916 and ash was the most abundant tree in the forest in 2011 (Fig. 2). However, low frequencies in the ground and shrub layers in 1935 and 1970 resulted in a decreasing density of trees with 10–19 cm dbh from 62 stems ha^{-1} in 1935 to only 10 stems ha^{-1} in 2011 (Table 2, see Appendix A: Table 3). This indicates a relatively weak regeneration during the period when tree canopies grew

Table 1. P -values from pair-wise Kolmogorov–Smirnov tests of diameter distributions from the four sampling occasions in Dalby Söderskog (1916, 1935, 1970 and 2011). For pairs of years within a tree species, numbers in bold are significant ($P < 0.05$).

Pair	<i>Ulmus</i>	<i>Fraxinus</i>	<i>Fagus</i>	<i>Quercus</i>
1916/1935	0.980	0.472	0.989	0.983
1916/1970	0.454	0.552	0.040	0.023
1916/2011	<0.001	0.739	0.693	0.001
1935/1970	0.910	0.100	0.237	0.130
1935/2011	<0.001	0.099	0.464	0.008
1970/2011	0.003	0.114	0.036	0.784

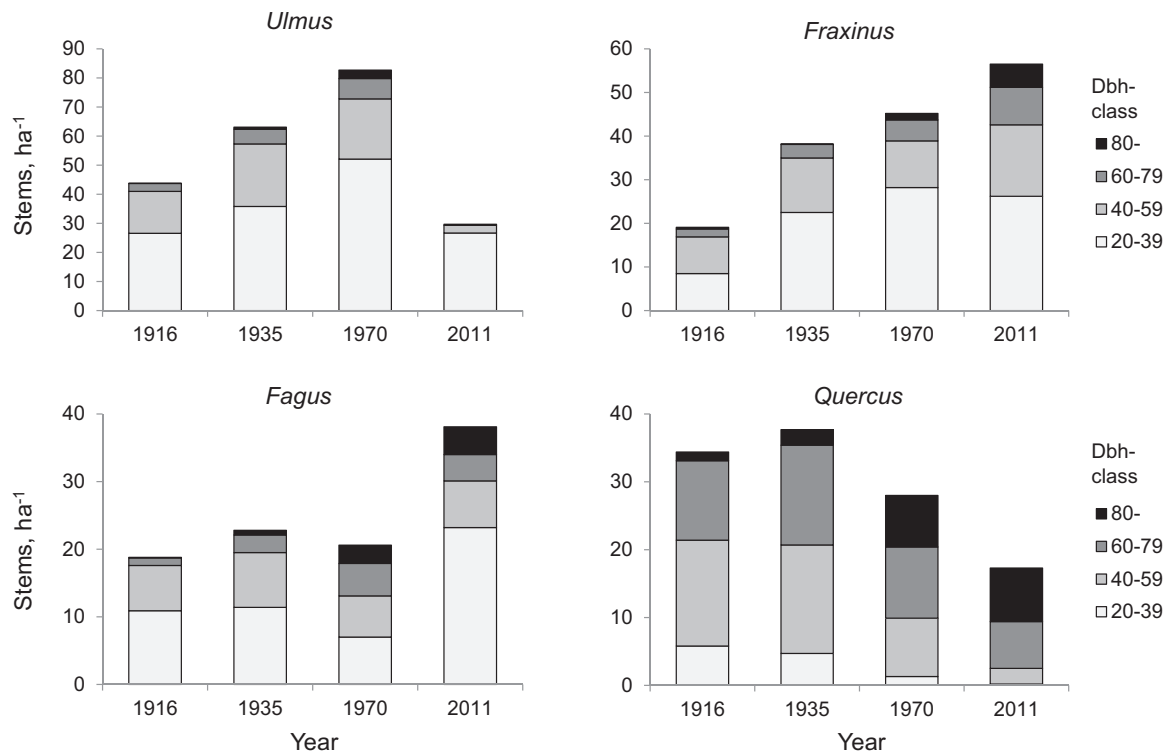


Fig. 3. Diameter distribution of the four main tree species in Dalby Söderskog during the period 1916–2011 (no. stems ha^{-1}). Only living trees with a dbh ≥ 20 cm at breast height are included. Note the different scales of the y-axis.

denser, thereby confirming hypothesis 1. More than half of the ash population having a dbh of ≥ 10 cm (56%) was classified as damaged by ash dieback, but only 3% of the ash trees in the sample plots were dead in 2011. The share of healthy ash trees tended to increase with dbh (Fig. 4, $R^2 = 0.58$, $P = 0.080$). The frequency of young ash increased significantly in both shrub layers between 1970 and 2011, and ash had replaced elm as the most frequent species in the lower shrub layer. In the ground layer, ash showed the largest increase in frequency between 1970 and 2011 and had a mean cover of 12% in 2011, which confirmed hypothesis 3 (Table 2).

Beech

The number of beech trees remained rather stable between 1909 and 1970, although the basal area steadily increased. This increase in basal area continued until 2011, but was then accompanied by a considerable increase in number of stems (Fig. 2). The dbh distributions show a relatively modest recruitment into the smaller dbh classes and a gradually ageing population until 1970 (Fig. 3), thereby not confirming hypothesis 1, and resulting in a significant change between 1916 and 1970 (Table 1). In 2011, however, the number of

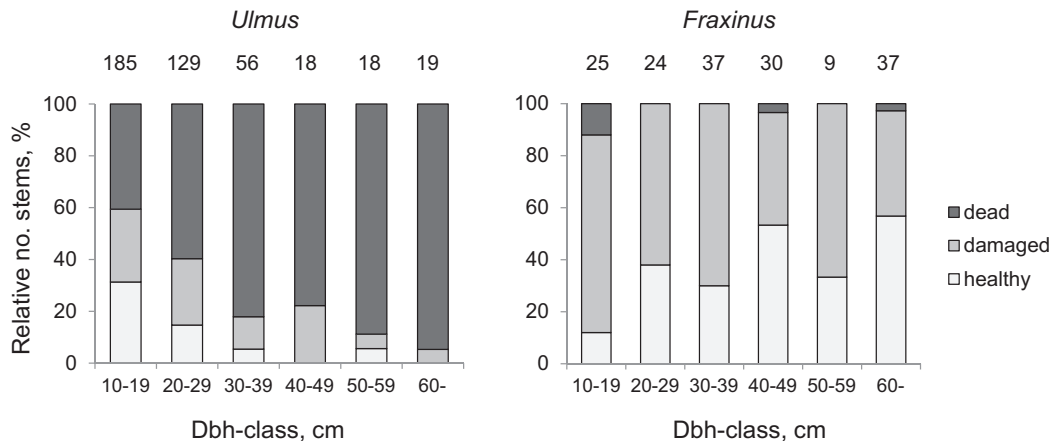


Fig. 4. Share (%) of standing *Ulmus glabra* and *Fraxinus excelsior* stems in Dalby Söderskog classified as dead, damaged by disease and healthy in 2011 in relation to dbh-class. Total number of surveyed stems is given above each bar.

Table 2. Frequency (%) of *Ulmus*, *Fraxinus*, *Fagus* and *Quercus* in the shrub (S) and ground (G) layers in 74 plots between 1935 and 2011. Statistically significant differences in frequency (χ^2 -test) between 1970 and 2011 are indicated by numbers in bold face in 2011 and the following significance levels: * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

	1935	1970	2011	<i>P</i>
S1 (2–8 m)				
<i>Ulmus glabra</i>	34	64	72	
<i>Fraxinus excelsior</i>	12	1	19	***
<i>Fagus sylvatica</i>	9	22	16	
<i>Quercus robur</i>	1	0	0	
S2 (1–2 m)				
<i>Ulmus glabra</i>	11	14	35	**
<i>Fraxinus excelsior</i>	1	5	54	***
<i>Fagus sylvatica</i>	4	4	16	*
<i>Quercus robur</i>	0	0	1	
G (<1 m)				
<i>Ulmus glabra</i>	19	3	5	
<i>Fraxinus excelsior</i>	9	11	84	***
<i>Fagus sylvatica</i>	7	4	34	***
<i>Quercus robur</i>	4	0	22	***

smaller beech (10–39 cm dbh) had increased considerably, confirming hypothesis 3 (Fig. 3, see Appendix A: Table 3). Beech also increased in frequency in the lower shrub layer and the ground layer between 1970 and 2011 (Table 2).

Oak

Between 1935 and 2011, the number of living oaks decreased linearly to 17 trees ha⁻¹ due to high mortality, but growth of surviving trees maintained a relatively constant basal area through the years (Fig. 2). The dbh distributions show the lack of oak regeneration since 1916 and the successive ageing of the population, which confirmed hypotheses 1 and 4 (Fig. 3). No change in size distribution took place between 1916 and 1935 while the distribution had changed significantly until the two latter inventories (Table 1). 120 oak stems were recorded as dead standing in 2011 (16% of the standing total 743 oak trees). We also recorded 270 downed oak stems and 17 old cut oak stumps, summing up to a total of 1030 oaks recorded in 2011. In addition, in total 455 young oaks (i.e. 12 trees ha⁻¹ with ≥ 1.3 m height and <10 cm dbh) were recorded in 2011, of which 323 were in the lower shrub layer (1.3–2 m height) and 132 in the upper shrub layer (2–8 m height). The frequency of oak regeneration also increased significantly in the ground layer between 1970 and 2011, which confirmed hypothesis 3 (Table 2).

Discussion

The succession in Dalby Söderskog after protection 1918 until the arrival of Dutch elm disease is typical for the

long-term development in mixed forest reserves in Europe and North America, where light-demanding oak species were replaced by more shade-tolerant trees, mainly as a result of lower levels of disturbance (Bernadzki, Bolibok, Brzezicki, Zajackowski, & Zybura 1998; Janik et al. 2008; Rogers, Rooney, Olson, & Waller 2008; Rohner, Bigler, Wunder, Brang, & Bugmann 2012; Wolf 2011). Our results are consistent with other studies which indicated that emerging dense understorey layers of shade-tolerant tree species are particularly important in hindering the survival and growth of oak seedlings (Aldrich, Parker, Romero-Severson, & Michler 2005; Götmark 2007; Lorimer, Chapman, & Lambert 1994; McDonald, Peet, & Urban 2003).

Our results also confirm case studies from England where continuous pressure from Dutch elm disease resulted in elm being restricted to the shrub layer (Lady Park Wood, Peterken & Mountford 1998), and from Denmark where gaps created by Dutch elm disease and other disturbance were filled by ash, which can be characterized as a fast growing gap specialist (Suserup forest, Emborg 1998). However, the current ash dieback has affected trees of all size classes, and the future role of this species in European forests seems highly uncertain (Pautasso et al. 2013).

Dutch elm disease and ash dieback may have far-reaching effects on associated forest biodiversity. For example, the bark of elm and ash provides a highly valuable habitat for many species of epiphytic bryophytes and lichens (Thor, Johansson, & Jönsson 2010; Pautasso et al. 2013). Hence, a fast decline of both species may result in local extinctions of epiphytes (Jönsson & Thor 2012). Epiphyte assemblages on elm, ash, beech and oak are, in general, more similar to each other than to those of other broadleaf trees (Ellis, Coppins, Eaton, & Simkin 2013). However, oak and beech trees may need to reach a higher age than elm and ash to develop a comparable bark habitat for epiphyte species of conservation concern (Fritz, Niklasson, & Churski 2009; Ranius, Johansson, Berg, & Niklasson 2008). In many forests, loss of elm and ash therefore will result in a substantial loss of suitable bark habitat for associated epiphyte species in the coming years.

Beech is the most abundant shade-tolerant tree species in European temperate forests and may be additionally favoured by negative effects of ash dieback in eutrophic forests, where ash usually shows a faster initial growth during regeneration (Emborg, Christensen, & Heilmann-Clausen 2000). However, beech is much more sensitive than ash and elm to high groundwater tables and flooding, which strongly limits its occurrence in floodplain forests and other moist eutrophic forests such as Dalby Söderskog (Glenz, Schlaepfer, Iorgulescu, & Kienast 2006). Hence, if European ash populations would be severely reduced by disease in the future, oak may be the only major tree species available to replace elm and ash in many forests (Glenz et al. 2006).

Pedunculate oak plays an outstanding role for biodiversity in European temperate forests, but has experienced a substantial historic decline, partly related to permanent

forest clearance for agriculture (Lindbladh & Foster 2010). Increasingly dense tree canopies in the 20th century in many remaining oak forests, including our study area and other forest reserves, have prevented regeneration and resulted in large gaps in the size distribution of oaks (Vera 2000). Pedunculate oak is the most important source of persistent stem cavities and hollows but usually does not develop hollows before an age of 150–200 years (Ranius, Niklasson, & Berg 2009). Therefore, apart from effects on epiphytes mentioned above, a long break in the continuity of old and large oaks will ultimately have severe negative effects for many species of invertebrates, birds and bats using tree hollows. Mortality within old oak populations is often caused by crown encroachment from shade-tolerant tree species (Drobyshev et al. 2008). Release from crown competition, either as an effect of tree diseases or active management, may prolong the life span of oaks and their function as habitat trees and as seed source for natural regeneration (Götmark 2009; Nordén, Paltto, Claesson, & Götmark 2012).

Successful natural oak regeneration in meso- and eutrophic forests depends on disturbance creating regeneration niches with sufficient light (Götmark 2007). In the past, such conditions were maintained by wood pasture management (Vera 2000), shifting agriculture systems (Bobiec, Kuijper, Niklasson, Romankiewicz, & Solecka 2011) or regular low intensity fires (Abrams 2003; McEwan et al. 2011). Moreover, flowering shrubs (e.g. Rosaceae) and forbs (e.g. Apiaceae), also associated with these land-uses and disturbance regimes, are essential for adult saproxylic beetles which have their larval development in wood of old trees. Open spaces now emerging from loss of elm and ash in forest reserves may be used to favour oak regeneration and important semi-open habitats once lost during succession (Heilmann-Clausen, Bruun, & Ejrnaes 2013). Our results show that natural regeneration of senescent oak populations is possible in such open spaces.

Although our data series on forest structure is one of the longest recorded, it is still far from encompassing a full development cycle in temperate forests (Emborg et al. 2000). Nevertheless it is long enough to evaluate the long-term effects of minimal intervention in oak rich forest reserves and of Dutch elm disease. We conclude that winners and losers change places as an effect of invasive pathogens, resulting in unexpected successions and both losses and gains in valuable ecological niches and habitat structures.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.baae.2014.02.002>.

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